

Beyond the Next Step for Latent-State Movement Models: Generative Validation Diagnostics for Hidden Markov Step-Selection Functions

Aurelien Nicosia

June 16, 2026

Scope of this draft

This draft focuses on the methodological core of the article: the HMM-SSF model formulation, the generative-validation logic, the diagnostic criteria, and the first simulation and empirical examples. The introduction, broader discussion, and conclusion are intentionally left for a later version, after the final simulation design and empirical emphasis have been fixed.

1 Extension of the SSF generative-validation framework

This manuscript is conceived as a latent-state extension of the generative validation framework for SSF and iSSF models developed in [Nicosia \(2026\)](#). The starting point is the same: a fitted step-selection model should not only describe local one-step decisions, but should also regenerate plausible trajectory-scale patterns when iterated forward. In the single-state setting, the validation target is the distribution of simulated trajectories generated by the fitted SSF or iSSF kernel. In the HMM-SSF setting, the validation target has two coupled components: the spatial trajectory and the latent regime sequence that governs state-specific movement and selection.

The original framework is organized around four trajectory-level pillars: emergent utilization distributions, mean squared displacement, path sinuosity, and barrier crossing. These pillars remain valid for HMM-SSFs because a multi-state model still generates an ordinary movement trajectory. The HMM extension adds a second diagnostic layer for the latent regime process: occupancy, residence times, switching rates, transition counts, and state-conditioned movement summaries. The goal is therefore not to replace the SSF diagnostics, but to decompose generative mismatch into spatial, movement-dynamic, and state-process components.

This framing also clarifies how the diagnostics should be interpreted. The Monte Carlo p-values are not generic measures of visual closeness across candidate models. Each fitted model generates its own reference distribution, with its own variability and sharpness. A model can therefore have a larger observed discrepancy but a less extreme p-value if its simulated trajectories are highly variable, whereas a sharper model can yield a smaller p-value for a smaller absolute discrepancy. The diagnostic question is model-specific: is the observed trajectory unusual under the generative distribution implied by that fitted model?

Table 1: Relationship between the original SSF generative-validation pillars and the HMM-SSF extension.

Original SSF pillar	Role retained for HMM-SSFs	Additional latent-state question
Emergent utilization distributions	Does the full fitted model re-generate the observed spatial footprint?	Are discrepancies driven by the state-specific SSF kernels or by unrealistic state occupancy?
Mean squared displacement	Does the model reproduce displacement scaling over time?	Does the latent switching process reproduce the temporal organization of movement regimes?
Path sinuosity	Does the model reproduce path-level tortuosity or directional persistence?	Are state-specific movement kernels and state durations jointly producing realistic path structure?
Barrier crossing	Does the model reproduce functional permeability of known barriers?	Does barrier response vary by state, and are barrier-crossing opportunities assigned to realistic regimes?

2 Model framework

Let $\{X_t\}_{t=1}^T$ denote a regular movement trajectory, with $X_t \in \mathbb{R}^2$, and let $\{S_t\}_{t=1}^T$ denote a latent behavioural regime process taking values in $\{1, \dots, K\}$. A hidden Markov step-selection function (HMM-SSF) combines a latent state process with a state-dependent movement and habitat-selection kernel. The central modelling assumption is that the animal alternates among a finite number of movement regimes, and that each regime has its own movement kernel and habitat-selection response.

Latent-state process. The latent process is specified through an initial distribution $\delta = (\delta_1, \dots, \delta_K)$ and transition probabilities

$$\Pr(S_{t+1} = j | S_t = i, z_t) = \Gamma_{ij}(z_t), \quad i, j \in \{1, \dots, K\},$$

where z_t denotes optional covariates affecting state persistence or switching. In the homogeneous case, $\Gamma_{ij}(z_t)$ reduces to a fixed transition matrix Γ . In a time-varying transition model, each row of $\Gamma(z_t)$ is usually parametrized by multinomial-logit contrasts. For example, using state i as the baseline destination from current state i , one may write

$$\log \left\{ \frac{\Gamma_{ij}(z_t)}{\Gamma_{ii}(z_t)} \right\} = \alpha_{ij} + \lambda_{ij}^\top z_t, \quad j \neq i.$$

This representation separates behavioural persistence from movement and habitat selection. The transition model controls when regimes are entered and left, whereas the state-dependent SSF controls how the animal moves and selects habitat while occupying a given regime.

State-dependent step-selection kernel. Conditional on the next state, the transition from X_t to X_{t+1} is described by a state-specific step-selection kernel. In its exponential form, a generic state-specific SSF can be written as

$$f_s(x | x_t, x_{t-1}, e_t) = \frac{q_s(x | x_t, x_{t-1}; \eta_s) \exp\{\beta_s^\top c(x, x_t, x_{t-1}, e_t)\}}{\int_{\Omega} q_s(u | x_t, x_{t-1}; \eta_s) \exp\{\beta_s^\top c(u, x_t, x_{t-1}, e_t)\} du}, \quad (1)$$

where q_s is the movement kernel in state s , η_s contains its movement parameters, e_t denotes environmental information available at time t , $c(\cdot)$ is a vector of movement and environmental covariates, and β_s is a state-specific selection coefficient vector. This formulation follows the general step-selection framework used to connect movement constraints and resource selection (Forester et al., 2009; Avgar et al., 2016) and its multi-state HMM-SSF extensions (Nicosia et al., 2017; Klappstein et al., 2023).

In applications fitted with used and available steps, the continuous normalizing integral in Eq. (1) is replaced by a discrete normalization over a candidate set $\mathcal{A}_t$ that contains the observed endpoint and a finite number of available endpoints. The state-specific probability assigned to candidate $a \in \mathcal{A}_t$ is then

$$p_s(a | \mathcal{A}_t) = \frac{q_s(a | x_t, x_{t-1}; \eta_s) \exp\{\beta_s^\top c(a, x_t, x_{t-1}, e_t)\}}{\sum_{u \in \mathcal{A}_t} q_s(u | x_t, x_{t-1}; \eta_s) \exp\{\beta_s^\top c(u, x_t, x_{t-1}, e_t)\}}. \quad (2)$$

This discrete form is useful for fitting because it connects the HMM-SSF to conditional logistic regression and to state-dependent conditional likelihood contributions.

Joint generative model. Conditioning on the first locations required to define step length and turning angle, the joint density of a latent state path and a movement path factorizes as

$$p_\theta(s_{1:T}, x_{1:T}) = \delta_{s_1} \prod_{t=1}^{T-1} \Gamma_{s_t s_{t+1}}(z_t) f_{s_{t+1}}(x_{t+1} | x_t, x_{t-1}, e_t), \quad (3)$$

up to the conditioning convention for the first step. The parameter vector θ contains the initial distribution, transition parameters, state-specific movement parameters, and state-specific selection coefficients. Equation (3) is the key theoretical object for this paper: it defines both the likelihood-based model and the simulation mechanism used for prediction and generative validation.

Likelihood and state uncertainty. Because the state path is latent, the observed-data likelihood sums over all possible state sequences. Let $E_t(s)$ denote the state-specific contribution of the observed step at time t under state s . In the discrete used-available form, $E_t(s)$ is given by Eq. (2) evaluated at the observed endpoint. The likelihood can then be evaluated with the standard HMM forward recursion (Rabiner, 1989; Langrock et al., 2012). With $P_t = \text{diag}\{E_t(1), \dots, E_t(K)\}$, a schematic likelihood expression is

$$L(\theta) = \delta^\top P_1 \Gamma(z_1) P_2 \Gamma(z_2) \cdots \Gamma(z_{T-1}) P_T \mathbf{1},$$

where $\mathbf{1}$ is a vector of ones. The exact indexing of P_t depends on whether the first step is treated as fixed or modelled explicitly, but the factorization is the same.

This likelihood representation matters for validation because state labels are not directly observed in empirical telemetry data. The fitted model provides filtered state probabilities, smoothed state probabilities, and a most likely Viterbi path. These summaries are useful, but they are not equivalent to observed regimes. Consequently, any diagnostic that uses state labels must be interpreted conditionally on a state-assignment rule or averaged over plausible state paths.

Prediction targets. A fitted HMM-SSF induces several distinct predictive objects. The one-step predictive distribution is a mixture over possible next states:

$$p_\theta(x_{t+1} | \mathcal{F}_t) = \sum_{i=1}^K \sum_{j=1}^K \alpha_t(i) \Gamma_{ij}(z_t) f_j(x_{t+1} | x_t, x_{t-1}, e_t), \quad (4)$$

where $\mathcal{F}_t$ denotes the information available up to time t , and $\alpha_t(i) = \Pr(S_t = i | \mathcal{F}_t)$ is the filtered probability of state i . Multi-step prediction is obtained by iterating the same mechanism: forecast the next state, draw or integrate over the next location, update the movement history and environmental covariates, and repeat. This iteration is what links local step-selection rules to emergent predictions such as long-term space use, displacement scaling, path structure, residence times, and barrier-crossing frequencies.

The distinction between local and generative prediction is central. A model may provide reasonable one-step conditional probabilities while producing unrealistic long-horizon trajectories once Eq. (3) is iterated. The diagnostics below are designed to evaluate this longer-horizon predictive distribution rather than only the local likelihood contributions.

3 Generative validation

The validation workflow uses the same Monte Carlo rank logic as the SSF framework in [Nicosia \(2026\)](#). It asks whether the observed trajectory, or a decoded state sequence derived from it, is unusual relative to replicate outputs generated by the fitted model. Let y denote the observed trajectory and let $Y^{(1)}, \dots, Y^{(M)}$ denote M simulated trajectories generated under a fitted HMM-SSF. For a diagnostic discrepancy D , the observed discrepancy is compared with a simulation reference distribution:

$$T_{\text{obs}} = \frac{1}{M} \sum_{m=1}^M D(y, Y^{(m)}),$$

$$T_m = \frac{1}{M-1} \sum_{\ell \neq m} D(Y^{(m)}, Y^{(\ell)}), \quad m = 1, \dots, M.$$

The Monte Carlo p-value is

$$p_{\text{MC}} = \frac{1 + \sum_{m=1}^M \mathbb{I}\{T_m \geq T_{\text{obs}}\}}{M + 1}.$$

For diagnostics where larger discrepancies indicate poorer agreement, small values of p_{MC} indicate that the observed trajectory or decoded state path is more discrepant than expected under the fitted model. The finite simulation resolution is $(M + 1)^{-1}$, which is reported with each diagnostic.

This rank-based construction is related to simulation-based model criticism and realized-discrepancy checks ([Gelman et al., 1996](#)) and to Monte Carlo significance testing ([Besag and Clifford, 1989](#)). The interpretation is conditional on the fitted model, the simulation strategy, and the chosen discrepancy. It is not a direct ranking of candidate models by visual distance, because each candidate model induces its own simulation reference distribution.

Compatibility alone is not sufficient for an informative diagnostic. A fitted model can pass because the simulated reference distribution is diffuse. The workflow therefore reports a sharpness score, computed from the concentration of simulated discrepancies. In the workflow used here, sharpness is a secondary label: it does not replace the Monte Carlo p-value, but it helps distinguish a concentrated compatibility check from one with low diagnostic resolution.

4 Diagnostic discrepancies for HMM-SSFs

The single-state generative-validation framework organizes model checking around biologically interpretable trajectory-level summaries: emergent space use, displacement dynamics, path structure, and barrier interactions. These criteria remain necessary for HMM-SSFs because the fitted model

still generates ordinary spatial trajectories. However, latent-state models introduce an additional process that must also be checked: the temporal organization of behavioural regimes. A multi-state model may reproduce the spatial footprint of the observed trajectory while assigning unrealistic time budgets to states, generating implausible residence times, or producing state-specific movement patterns that do not match the decoded regimes.

We therefore separate the diagnostics into two complementary groups. The first group applies the original trajectory-level checks to the full simulated path, without conditioning on state labels. The second group targets the latent regime process and asks whether the fitted model regenerates plausible occupancy, persistence, switching structure, and state-conditioned movement. These diagnostics are not combined into a single scalar goodness-of-fit measure. As in the one-state framework, they are intended to form a diagnostic dashboard in which each criterion probes a distinct way in which the fitted model can fail to generate realistic movement.

Table 2: Overview of generative diagnostics for HMM-SSFs. The first four rows extend the trajectory-level pillars of the single-state SSF framework; the remaining rows target the latent state process and state-conditioned movement.

Diagnostic family		Target pattern	Summary statistic	Typical failure modes
Emergent utilization distributions		Long-term spatial configuration of space use.	Spatial discrepancy between observed and simulated utilization distributions.	Spatial leakage, over-diffusion, or failure to reproduce concentration.
Mean squared displacement		Temporal displacement scaling across lags.	Integrated discrepancy between observed and simulated MSD curves.	Over-diffusion, under-diffusion, or missing multi-state temporal structure.
Path sinuosity		Path-level tortuosity and directional persistence.	Absolute discrepancy in straightness index.	Paths too straight, too tortuous, or behaviourally unrealistic.
Barrier crossing		Functional connectivity across known linear features.	Observed barrier crossings relative to simulated crossings.	Overestimated permeability or state-specific barrier response mismatch.
State occupancy		Time budget allocated to each latent regime.	Per-state occupancy and global total-variation discrepancy.	Dominant states, missing states, or unrealistic regime allocation.
Residence time		Persistence and dwell-time shape within regimes.	Residence-time distribution discrepancy and geometric dwell-time check.	Non-geometric residence times, memory, or inadequate Markov persistence.
Switching and transitions		Frequency and direction of transitions among regimes.	Switching rate and normalized transition-count discrepancy.	Incorrect switching frequency or misspecified time-varying transition model.
State-conditioned movement		Movement phenotype within each decoded or simulated regime.	State-specific step-length, turning-angle, and MSD discrepancies.	Misspecified state-specific movement kernels or poorly separated regimes.

4.1 Trajectory-level diagnostics

Trajectory-level diagnostics do not require a latent state path. They evaluate emergent movement patterns produced by the full fitted model, and correspond to the original SSF generative-validation pillars. Let $X_{1:T}^{(0)}$ denote the observed trajectory and $X_{1:T}^{(m)}$, $m = 1, \dots, M$, denote simulated trajectories generated under a given HMM-SSF simulation strategy. These diagnostics intentionally ignore the state labels after simulation: they ask whether the fitted model, after integrating over its latent regime process, produces realistic spatial paths. This distinction is important because a model can have interpretable latent states while still producing unrealistic long-horizon trajectories, or can generate plausible spatial paths while allocating movement to unrealistic regimes.

Utilization-distribution discrepancy. This diagnostic targets emergent space use, as in the original SSF framework. For each trajectory, we construct an empirical utilization distribution (UD) from the visited locations. Let μ_0 denote the empirical UD of the observed trajectory and let μ_m denote the empirical UD of simulated trajectory m . In the full spatial version, mismatch between UD's can be measured by the 1-Wasserstein distance, which quantifies the minimal amount of spatial “work” required to transport probability mass from one distribution to another (Villani, 2009). In the present HMM-SSF analyses, we use a computationally light approximation based on the sum of one-dimensional Wasserstein distances between marginal coordinate distributions:

$$D_{\text{UD}}(a, b) = W_1(a_x, b_x) + W_1(a_y, b_y),$$

where a_x and b_x are the empirical distributions of the x coordinates, and a_y and b_y are the corresponding empirical distributions of the y coordinates. The observed discrepancy is

$$T_{\text{obs}}^{\text{UD}} = \frac{1}{M} \sum_{m=1}^M D_{\text{UD}}(\mu_0, \mu_m),$$

and the simulated reference discrepancies are computed by leave-one-out comparison among simulated trajectories:

$$T_m^{\text{UD}} = \frac{1}{M-1} \sum_{\ell \neq m} D_{\text{UD}}(\mu_m, \mu_\ell).$$

Under a well-specified generative model, the observed UD should not be more spatially isolated from the simulated UD's than simulated UD's are from one another. A small Monte Carlo p-value therefore indicates that the fitted model does not reproduce the long-run spatial footprint of the observed trajectory. In an HMM-SSF, this failure can arise from the state-specific SSF kernels, from unrealistic state occupancy or residence near particular habitats, or from an interaction between movement and switching dynamics.

Mean squared displacement discrepancy. Mean squared displacement (MSD) targets the temporal scaling of movement (Turchin, 1998), and is closely related to net squared displacement summaries used to characterize movement phenotypes (Bastille-Rousseau et al., 2016). For trajectory $r \in \{0, 1, \dots, M\}$ and lag k , define

$$\text{MSD}_r(k) = \frac{1}{T-k} \sum_{t=1}^{T-k} \|X_{t+k}^{(r)} - X_t^{(r)}\|^2.$$

Let $\mathcal{K}$ be the set of lags used for the diagnostic, and let $w_k \geq 0$ be optional weights used to emphasize particular temporal scales (with $w_k = 1$ in the unweighted case). The model-expected MSD curve is estimated from the simulated ensemble,

$$\overline{\text{MSD}}_{\text{sim}}(k) = \frac{1}{M} \sum_{m=1}^M \text{MSD}_m(k).$$

The observed MSD discrepancy is then

$$T_{\text{obs}}^{\text{MSD}} = \sum_{k \in \mathcal{K}} w_k [\text{MSD}_0(k) - \overline{\text{MSD}}_{\text{sim}}(k)]^2. \quad (5)$$

For each simulated trajectory m , the reference statistic is computed against the leave-one-out simulated mean,

$$T_m^{\text{MSD}} = \sum_{k \in \mathcal{K}} w_k [\text{MSD}_m(k) - \overline{\text{MSD}}_{-m}(k)]^2, \quad (6)$$

where

$$\overline{\text{MSD}}_{-m}(k) = \frac{1}{M-1} \sum_{\ell \neq m} \text{MSD}_{\ell}(k).$$

This discrepancy is sensitive to both the magnitude and the shape of the MSD curve. It can detect trajectories that diffuse too rapidly, remain too spatially constrained, or exhibit non-smooth displacement dynamics that are not reproduced by the fitted HMM-SSF. In multi-state models, MSD is particularly useful because it depends jointly on state-specific movement kernels and on the temporal arrangement of states. A model may match average step lengths within states but still fail to reproduce MSD if residence times or switching patterns are wrong.

Sinuosity discrepancy. Path sinuosity targets the geometry of the realized path rather than its spatial footprint or displacement scaling. We use the straightness index (Benhamou, 2004),

$$\text{SI}_r = \frac{\|X_T^{(r)} - X_1^{(r)}\|}{\sum_{t=1}^{T-1} \|X_{t+1}^{(r)} - X_t^{(r)}\|}, \quad 0 \leq \text{SI}_r \leq 1.$$

Low values correspond to more tortuous paths, whereas values close to one indicate comparatively straight movement. Let

$$\overline{\text{SI}}_{\text{sim}} = \frac{1}{M} \sum_{m=1}^M \text{SI}_m$$

be the simulated mean straightness index. The observed discrepancy is

$$T_{\text{obs}}^{\text{SI}} = |\text{SI}_0 - \overline{\text{SI}}_{\text{sim}}|,$$

and the simulated reference discrepancies are

$$T_m^{\text{SI}} = |\text{SI}_m - \overline{\text{SI}}_{-m}|, \quad \overline{\text{SI}}_{-m} = \frac{1}{M-1} \sum_{\ell \neq m} \text{SI}_{\ell}.$$

The absolute discrepancy is used because both directions are biologically meaningful: simulated paths can be too straight, suggesting excessive directional persistence or too much time in travelling-like regimes, or too tortuous, suggesting excessive turning or too much time in localized regimes. In HMM-SSFs, sinuosity is therefore a useful bridge between movement kernels and state occupancy: it can flag cases where the marginal spatial distribution looks acceptable but the behavioural geometry of the path is unrealistic.

Barrier-crossing discrepancy. When a known barrier geometry is available, the HMM-SSF workflow inherits the barrier-crossing pillar from the single-state SSF framework. The diagnostic compares the observed number of intersections between movement steps and a specified linear feature with the corresponding simulated distribution. Let B denote the barrier geometry and let

$$C(X_{1:T}^{(r)}; B) = \sum_{t=1}^{T-1} \mathbb{I}\{[X_t^{(r)}, X_{t+1}^{(r)}] \cap B \neq \emptyset\}$$

be the number of step segments crossing the barrier. When the scientific concern is excess permeability, the one-sided Monte Carlo p-value is

$$p_{\text{barrier}} = \frac{1 + \sum_{m=1}^M \mathbb{I}\{C(X_{1:T}^{(m)}; B) \leq C(X_{1:T}^{(0)}; B)\}}{M + 1}. \quad (7)$$

Small values indicate that the observed trajectory crosses the barrier less often than expected under the fitted model, suggesting that the model overestimates permeability. If the ecological concern is instead that the model is too restrictive, the inequality in Eq. (7) can be reversed. In a latent-state model, barrier mismatch can arise from the state-specific selection kernels, from unrealistic residence near the barrier, or from the state process assigning barrier-crossing opportunities to the wrong regimes. This diagnostic is therefore interpreted jointly with state occupancy, residence time, and state-conditioned movement summaries when state-specific barrier response is scientifically relevant. As in the single-state framework, the barrier must be specified a priori and the diagnostic can have low power when crossing events are rare.

4.2 State-process diagnostics

State-process diagnostics require an observed or decoded state path and a corresponding simulated state path. In empirical applications, observed states are typically decoded, for example using the Viterbi path, rather than directly observed. The diagnostics should therefore be interpreted as conditional on a state-assignment procedure, or evaluated across several state-simulation strategies to assess sensitivity to latent-state uncertainty. Let $A_{1:T}^{(0)}$ denote the decoded or supplied state path associated with the observed trajectory, and let $A_{1:T}^{(m)}$, $m = 1, \dots, M$, denote simulated state paths generated under the fitted HMM-SSF. The following diagnostics ask whether $A_{1:T}^{(0)}$ is typical of the state-process distribution induced by the fitted model, either marginally or after conditioning on the chosen state-assignment strategy.

State occupancy. State occupancy measures the time budget allocated to each behavioural regime. For a state sequence $a_{1:T}$, define the occupancy proportion of state s as

$$\pi_s(a) = \frac{1}{T} \sum_{t=1}^T \mathbb{I}\{a_t = s\}.$$

For each state, the observed occupancy discrepancy is the absolute difference between the decoded observed occupancy and the mean simulated occupancy:

$$T_{\text{obs},s}^{\text{occ}} = \left| \pi_s(A^{(0)}) - \frac{1}{M} \sum_{m=1}^M \pi_s(A^{(m)}) \right|.$$

The simulated reference discrepancies are obtained by leave-one-out comparison:

$$T_{m,s}^{\text{occ}} = \left| \pi_s(A^{(m)}) - \frac{1}{M-1} \sum_{\ell \neq m} \pi_s(A^{(\ell)}) \right|.$$

A global occupancy summary is also useful:

$$T_{\text{obs}}^{\text{occ,global}} = \frac{1}{K} \sum_{s=1}^K \left| \pi_s(A^{(0)}) - \frac{1}{M} \sum_{m=1}^M \pi_s(A^{(m)}) \right|.$$

For two-state models, per-state absolute discrepancies can be identical because the two occupancy proportions are complementary. In such cases, the global occupancy statistic provides the model-level interpretation, while per-state rows identify which biological regimes are over- or under-represented.

Residence time. Residence times are run lengths of consecutive visits to the same state. Let

$$\mathcal{R}_s(a) = \{r_{s,1}(a), \dots, r_{s,n_s(a)}(a)\}$$

be the empirical distribution of run lengths in state s for sequence a . The simulation-based residence-time diagnostic compares $\mathcal{R}_s(A^{(0)})$ with the corresponding simulated distributions using a one-dimensional Wasserstein distance:

$$T_{\text{obs},s}^{\text{res}} = \frac{1}{M} \sum_{m=1}^M W_1\{\mathcal{R}_s(A^{(0)}), \mathcal{R}_s(A^{(m)})\}. \quad (8)$$

The simulated reference statistic is

$$T_{m,s}^{\text{res}} = \frac{1}{M-1} \sum_{\ell \neq m} W_1\{\mathcal{R}_s(A^{(m)}), \mathcal{R}_s(A^{(\ell)})\}. \quad (9)$$

A small p-value indicates that residence times in the decoded observed state path are unusual relative to residence times generated by the fitted transition model. This is a stronger statement than an occupancy discrepancy: a model can reproduce the overall time budget in each state while producing bouts that are too short, too long, or too variable.

Geometric residence-time check. In a homogeneous HMM, the residence time in state s has a geometric distribution with leaving probability $1 - \Gamma_{ss}$:

$$\Pr(R_s = r) = \Gamma_{ss}^{r-1} (1 - \Gamma_{ss}), \quad r = 1, 2, \dots \quad (10)$$

This property follows directly from the Markov assumption: once the process is in state s , the probability of leaving at the next time step is constant and does not depend on how long the process has already remained in that state. The geometric residence-time diagnostic targets this assumption directly. For each state, bootstrap samples $G_s^{(1)}, \dots, G_s^{(B)}$ are generated from the geometric distribution in Eq. (10), with sample size matching the number of observed residence bouts in state s . The observed discrepancy is

$$T_{\text{obs},s}^{\text{geo}} = \frac{1}{B} \sum_{b=1}^B W_1\{\mathcal{R}_s(A^{(0)}), G_s^{(b)}\},$$

and the parametric-bootstrap reference uses leave-one-out distances among the geometric samples:

$$T_{b,s}^{\text{geo}} = \frac{1}{B-1} \sum_{c \neq b} W_1(G_s^{(b)}, G_s^{(c)}).$$

A small p-value indicates that the decoded dwell-time distribution is unusual under the geometric distribution implied by the fitted homogeneous HMM. This is evidence of a residence-time mismatch, not definitive proof that a semi-Markov model is the correct alternative. The diagnostic is defined only for homogeneous transition matrices; for time-varying transition probabilities, residence-time adequacy should be assessed using model-generated state paths as in Eqs. (8)–(9).

Switching rate and transition counts. The switching rate summarizes the overall frequency of changes between latent regimes:

$$\rho(a) = \frac{1}{T-1} \sum_{t=1}^{T-1} \mathbb{I}\{a_{t+1} \neq a_t\}.$$

The observed switching discrepancy is

$$T_{\text{obs}}^{\text{switch}} = \left| \rho(A^{(0)}) - \frac{1}{M} \sum_{m=1}^M \rho(A^{(m)}) \right|,$$

with leave-one-out simulated discrepancies defined analogously. This diagnostic detects whether the fitted model switches too often or not often enough on average, but it does not identify which transitions are responsible.

To retain directional information, let

$$N_{ij}(a) = \sum_{t=1}^{T-1} \mathbb{I}\{a_t = i, a_{t+1} = j\}$$

be the empirical transition-count matrix, and define its normalized version

$$P_{ij}(a) = \frac{N_{ij}(a)}{\sum_{u=1}^K \sum_{v=1}^K N_{uv}(a)}.$$

Writing $\text{vec}\{P(a)\}$ for the vectorized normalized transition matrix, the transition-count discrepancy is

$$T_{\text{obs}}^{\text{trans}} = \left\| \text{vec}\{P(A^{(0)})\} - \frac{1}{M} \sum_{m=1}^M \text{vec}\{P(A^{(m)})\} \right\|_2,$$

with simulated reference discrepancies computed by leave-one-out replacement of the simulated mean. Transition-count diagnostics can agree with the fitted HMM even when residence-time shape is misspecified, because a Markov chain can reproduce average transition frequencies without reproducing non-geometric dwell-time structure.

State-conditioned movement summaries. State-conditioned diagnostics compare movement summaries within each decoded or simulated state. We consider state-conditioned mean squared displacement, step-length distributions, and turning-angle summaries. These diagnostics target a different failure mode from occupancy or residence time: they ask whether the state-specific

movement kernels reproduce the movement phenotype associated with each regime, conditional on the chosen state assignment.

For step lengths, steps are assigned to the state at the arrival location. Let $L_s^{(r)}$ denote the empirical distribution of step lengths assigned to state s in trajectory r . The state-specific step-length discrepancy is

$$T_{\text{obs},s}^{\text{step}} = \frac{1}{M} \sum_{m=1}^M W_1(L_s^{(0)}, L_s^{(m)}),$$

with leave-one-out Wasserstein distances among simulated state-specific step-length distributions used as the reference. A small p-value indicates that the movement kernel for state s does not reproduce the observed state-specific step-length distribution.

For turning angles, we summarize directional concentration in state s using the mean resultant length

$$R_s = \sqrt{\left[\frac{1}{n_s} \sum_{q=1}^{n_s} \cos(\alpha_{s,q}) \right]^2 + \left[\frac{1}{n_s} \sum_{q=1}^{n_s} \sin(\alpha_{s,q}) \right]^2},$$

where $\alpha_{s,q}$ are the turning angles assigned to state s . Values near zero indicate dispersed turning angles, while values near one indicate strong directional persistence. The diagnostic compares the observed R_s with the mean simulated R_s and uses leave-one-out simulated discrepancies as the reference.

For within-state MSD, locations assigned to state s are extracted in temporal order and treated as a state-specific subsequence. An MSD curve is then computed on this ordered subsequence and compared with the mean simulated state-specific MSD curve using the same integrated squared discrepancy as in Eqs. (5)–(6). This diagnostic is descriptive, because the state-specific subsequence need not have evenly spaced dwell-time structure after extraction, but it is useful for identifying regimes whose within-state displacement dynamics are not reproduced by the fitted movement kernel.

5 Modelling and prediction with HMM-SSFs

The generative representation in Eq. (3) makes an HMM-SSF useful not only for retrospective behavioural interpretation, but also for prediction. Prediction is not a single task, however. Depending on the scientific question, the target may be the next step, the next behavioural state, a full future trajectory, an emergent utilization distribution, a barrier-crossing frequency, or a counterfactual response to a changed landscape. These targets use the same fitted model but place different demands on the transition model, the state-specific movement kernels, and the state-specific selection functions.

5.1 From fitted model to predictive distribution

At short horizons, prediction can be expressed analytically as mixtures of state-specific kernels. Eq. (4) shows that the one-step predictive distribution combines three elements: the current state uncertainty α_t , the transition probabilities $\Gamma(z_t)$, and the state-specific SSF kernels f_j . This decomposition is useful for interpretation. A predicted move can be uncertain because the next behavioural state is uncertain, because the state-specific movement kernel is diffuse, or because habitat selection assigns similar weights to many available endpoints.

At longer horizons, exact integration is generally impractical because future locations determine future movement covariates, available steps, and environmental values. Forward simulation is

Table 3: Prediction targets induced by a fitted HMM-SSF.

Prediction target	Model component emphasized	Typical output
One-step movement	Filtered state probabilities and state-specific SSF kernels	Conditional surface or candidate-step probabilities for X_{t+1}
State forecasting	Transition matrix or transition model $\Gamma(z_t)$	Predicted state probabilities, occupancy, switching probability
Trajectory forecasting	Full latent-state and movement process	Simulated paths over a specified horizon
Emergent space use	Iterated movement process and landscape covariates	Predicted utilization distribution or occupancy surface
Connectivity or barrier response	State-specific selection and movement near known barriers	Crossing counts, first-passage summaries, side-specific residence
Scenario prediction	Modified covariates or transition drivers	Counterfactual trajectories and derived summaries

therefore the natural predictive tool. A single predictive replicate is generated by repeating the following steps: sample the next state from $\Gamma(z_t)$, sample the next location from the corresponding state-specific SSF, update the movement history, update any time-varying covariates, and continue until the prediction horizon is reached. An ensemble of such replicates approximates the predictive distribution of any trajectory-level summary.

5.2 Latent-state simulation as predictive conditioning

The hidden component of an HMM-SSF is a discrete stochastic process. Simulating it means generating a state sequence

$$S_{1:H}^* = (S_1^*, \dots, S_H^*)$$

over a prediction or validation horizon H , then using this sequence to choose which state-specific SSF kernel generates each movement step. For the full generative model, the simulated state path is drawn from the fitted initial distribution and transition model:

$$S_1^* \sim \text{Categorical}(\delta),$$

$$S_{h+1}^* | S_h^* = i, z_h^* \sim \text{Categorical}\{\Gamma_{i1}(z_h^*), \dots, \Gamma_{iK}(z_h^*)\}, \quad h = 1, \dots, H-1.$$

Equivalently, if $U_h \sim \text{Uniform}(0, 1)$, then

$$S_{h+1}^* = \min \left\{ j : \sum_{\ell=1}^j \Gamma_{i\ell}(z_h^*) \geq U_h \right\}, \quad i = S_h^*.$$

This inverse-transform expression is useful because it makes clear that the next hidden state is sampled from the row of the transition matrix corresponding to the current hidden state.

In a homogeneous HMM, $\Gamma(z_h^*) = \Gamma$ for all h , so the hidden state simulation is an ordinary finite-state Markov chain. In a time-varying HMM, the transition matrix is recomputed at each step from the relevant covariates:

$$\Gamma(z_h^*) = [\Gamma_{ij}(z_h^*)]_{i,j=1}^K.$$

These covariates may be external, such as time of day or season, or they may be functions of the simulated trajectory itself. When transition covariates depend on the simulated trajectory, the hidden

and spatial components must be updated jointly: draw S_{h+1}^* , draw X_{h+1}^* from the corresponding state-specific SSF, update the covariates, then proceed to the next step.

Conditional on the simulated hidden path, the simulated spatial path satisfies

$$X_{h+1}^* | X_h^*, X_{h-1}^*, S_{h+1}^* = s \sim f_s(\cdot | X_h^*, X_{h-1}^*, e_h^*).$$

Thus the full predictive simulation can be written as the product

$$p_\theta(s_{1:H}^*, x_{1:H}^*) = p_\theta(s_{1:H}^*) p_\theta(x_{1:H}^* | s_{1:H}^*),$$

where the first term is the hidden Markov chain and the second term is the sequence of state-conditioned SSF draws. This decomposition is the basis for the different simulation strategies below.

We distinguish four simulation strategies. They are not competing fitted models; they are four ways of conditioning the same fitted HMM-SSF before simulating the spatial trajectory. The difference among them is entirely in how the latent path $S_{1:H}^*$ is generated.

Markov generation

Markov generation is the full generative use of the fitted HMM-SSF. The latent state sequence is simulated from the fitted initial distribution and fitted transition model:

$$\begin{aligned} S_1^* &\sim \text{Categorical}(\hat{\delta}), \\ S_{h+1}^* | S_h^* = i &\sim \text{Categorical}\{\hat{\Gamma}_{i1}(z_h^*), \dots, \hat{\Gamma}_{iK}(z_h^*)\}. \end{aligned}$$

The spatial path is then generated conditionally on this sampled state path:

$$X_{h+1}^* | X_h^*, X_{h-1}^*, S_{h+1}^* = s \sim f_s(\cdot | X_h^*, X_{h-1}^*, e_h^*; \hat{\theta}).$$

Equivalently, this strategy samples from

$$p_{\hat{\theta}}(s_{1:H}^*, x_{1:H}^*) = p_{\hat{\theta}}(s_{1:H}^*) p_{\hat{\theta}}(x_{1:H}^* | s_{1:H}^*).$$

It is the appropriate strategy for genuine forward prediction, simulation of space use, and scenario analysis because it propagates both latent-state uncertainty and movement stochasticity.

Viterbi-conditioned generation

Viterbi-conditioned generation fixes the latent path to the most likely decoded state sequence for the observed trajectory. If

$$\hat{s}_{1:T}^V = \arg \max_{s_{1:T}} p_{\hat{\theta}}(s_{1:T} | x_{1:T}),$$

then this strategy sets

$$S_{1:T}^* = \hat{s}_{1:T}^V$$

and simulates only the spatial component

$$X_{1:T}^* \sim p_{\hat{\theta}}(X_{1:T} | S_{1:T} = \hat{s}_{1:T}^V).$$

This is not a full predictive distribution for future movement, because the latent state sequence is treated as known. Its purpose is diagnostic: it asks whether the state-specific movement and selection kernels can reproduce the trajectory when the regime sequence is held fixed. If the Viterbi-conditioned strategy is compatible with the observed trajectory but Markov generation is not, the discrepancy is more plausibly associated with the transition model, state occupancy, residence times, or switching structure.

Posterior state-path generation

Posterior state-path generation samples the latent path from the smoothing distribution implied by the fitted model and observed trajectory:

$$S_{1:T}^* \sim p_{\hat{\theta}}(S_{1:T} | X_{1:T} = x_{1:T}).$$

In practice, this can be implemented by forward-filtering backward-sampling. If $\alpha_h(i) = \Pr_{\hat{\theta}}(S_h = i | X_{1:h} = x_{1:h})$ denotes the filtered probability at time h , then a backward draw satisfies

$$S_T^* \sim \Pr_{\hat{\theta}}(S_T | X_{1:T} = x_{1:T}),$$

$$\Pr(S_h^* = i | S_{h+1}^* = j, X_{1:T} = x_{1:T}) \propto \alpha_h(i) \hat{\Gamma}_{ij}(z_h), \quad h = T-1, \dots, 1.$$

After a posterior state path is sampled, the spatial trajectory is simulated from

$$X_{1:T}^* \sim p_{\hat{\theta}}(X_{1:T} | S_{1:T}^*).$$

This strategy incorporates uncertainty in state assignment while remaining conditioned on the information in the observed trajectory. It is useful when the scientific question is whether diagnostic conclusions depend on treating a single decoded state path as fixed.

Viterbi-tube generation

Viterbi-tube generation is a sensitivity strategy centred on the Viterbi path. Let $d(s_{1:T}, \hat{s}_{1:T}^V)$ be a distance between state sequences, for example a Hamming distance or another application-specific discrepancy. For a tolerance ϵ , define the tube around the Viterbi path as

$$\mathcal{T}_{\epsilon}(\hat{s}^V) = \{s_{1:T} : d(s_{1:T}, \hat{s}_{1:T}^V) \leq \epsilon\}.$$

The idealized Viterbi-tube strategy samples from the posterior distribution restricted to this tube:

$$\Pr_{\hat{\theta}}(S_{1:T}^* = s_{1:T} | X_{1:T} = x_{1:T}, S_{1:T} \in \mathcal{T}_{\epsilon}) = \frac{p_{\hat{\theta}}(s_{1:T} | x_{1:T}) \mathbb{I}\{s_{1:T} \in \mathcal{T}_{\epsilon}(\hat{s}^V)\}}{\sum_{\tilde{s}_{1:T} \in \mathcal{T}_{\epsilon}(\hat{s}^V)} p_{\hat{\theta}}(\tilde{s}_{1:T} | x_{1:T})}.$$

The spatial trajectory is then simulated conditionally on the sampled path. This strategy is not intended as a new ecological model. It asks whether the diagnostic conclusions depend on one exact Viterbi sequence or remain stable for nearby high-support regime sequences.

5.3 Prediction after model checking

The diagnostics in this paper are intended to make prediction more explicit. For example, if utilization-distribution diagnostics are compatible but MSD diagnostics fail, the model may still be useful for long-term spatial footprint prediction while being unreliable for transient displacement or encounter-rate prediction. If state occupancy is compatible but residence-time diagnostics fail, predicted time budgets may be reasonable while predicted bout durations or switching sequences remain unreliable. If state-conditioned step-length diagnostics fail, forecasts that depend on travel distance, connectivity, or barrier-crossing opportunities should be treated cautiously even if global space-use summaries look acceptable.

This perspective separates model fitting from model use. A fitted HMM-SSF can be statistically estimable and biologically interpretable without being adequate for every predictive purpose. Generative validation identifies which features of the predictive distribution are supported by the fitted model and which features require model refinement, additional states, time-varying transition structure, memory, or richer state-specific movement kernels.

6 Simulation study

This section defines the proposed simulation programme. No simulation results are reported at this stage. The objective is to specify a design that can be discussed before implementation, so that the final study tests the methodological claims of the paper rather than only illustrating the software.

The simulation study has four goals. First, it should assess calibration under well-specified HMM-SSFs: when the fitted model matches the data-generating process, the observed trajectory should not be systematically extreme relative to the fitted generative distribution. Second, it should evaluate diagnostic sensitivity under controlled misspecification, with each scenario targeting a distinct model component. Third, it should determine whether trajectory-level and state-process diagnostics identify different failure modes. Fourth, it should clarify how conclusions depend on the way latent state uncertainty is handled during simulation.

General protocol. All scenarios will use the same basic workflow. A known data-generating process will be used to simulate a movement trajectory, the candidate model or models will be fitted to that trajectory, and the fitted model will be used to generate replicated trajectories and state sequences. The diagnostic workflow will then compare the observed simulated trajectory with the model-generated replicates. For each scenario, the planned outputs are Monte Carlo p-values, observed discrepancies, simulation-reference variability, diagnostic sharpness, and rejection frequencies across independent replicate data sets.

The default synthetic landscape will contain one smooth habitat covariate and, for the barrier scenarios, one linear barrier. The default behavioural interpretation will use an encamped state with short, tortuous steps and a travelling state with longer, more persistent movement. Step lengths will be generated from state-specific positive distributions, turning angles from state-specific circular distributions, and endpoint weights from state-specific selection coefficients. Unless a scenario explicitly changes one component, the fitted model will contain the same covariates and the same number of states as the data-generating process.

Table 4: Planned simulation scenarios for evaluating HMM-SSF generative diagnostics.

ID	Purpose	Fitted models	Primary diagnostic target
S0	Calibration under a correct two-state HMM-SSF.	Correct two-state HMM-SSF.	Approximate calibration of all diagnostic p-values.
S1	Loss induced by collapsing two behavioural regimes into one.	One-state SSF and correct two-state HMM-SSF.	Trajectory-level MSD and sinuosity, with UD used as a contrasting spatial diagnostic.
S2	Incorrect number of latent states.	One-, two-, three-, and four-state HMM-SSFs fitted to data generated with three states.	Occupancy, state-conditioned movement, and trajectory geometry.
S3	Missing time variation in the transition matrix.	Fixed-TPM and time-varying-TPM HMM-SSFs.	Occupancy, switching rate, residence time, and MSD.
S4	Misspecified state-specific movement kernel.	Simplified movement kernel compared with the correct movement kernel.	State-specific step length, turning angle, within-state MSD, and global MSD.
S5	Misspecified habitat-selection structure.	No habitat effect, shared habitat effect, and correct state-specific habitat effect.	UD discrepancy and state-conditioned movement or space-use summaries.
S6	Misspecified barrier response.	No barrier response, common barrier response, and correct state-specific barrier response.	Barrier crossing and state-specific crossing opportunities.
S7	Non-geometric residence times.	HMM-SSF fitted to trajectories generated from an HSMM-like state process.	Residence-time and geometric-dwell diagnostics, with occupancy as a negative control.
S8	Sensitivity to state uncertainty and weak state separation.	Correct HMM-SSF analyzed with Markov, Viterbi-conditioned, posterior path, and Viterbi-tube simulations.	Robustness of state-process diagnostics to the state-assignment rule.

Scenario S0: well-specified two-state HMM-SSF. The data-generating process will be a two-state HMM-SSF with one encamped state and one travelling state. The two states will differ in step-length scale, turning-angle concentration, habitat-selection strength, and persistence. The fitted model will use the correct number of states, the correct habitat covariate, and the correct transition structure. This scenario is the calibration check for the entire workflow. The expected result is that trajectory-level and state-process p-values are not systematically small, apart from finite-sample and state-decoding effects.

Scenario S1: single-state approximation to a two-state process. The data-generating process will be the same as in S0, but the fitted comparison will include a one-state SSF and the correctly specified two-state HMM-SSF. This scenario tests the main motivation for the HMM extension of the single-state framework. The one-state model may reproduce parts of the spatial footprint, especially when the two states use similar areas, but it should struggle to reproduce displacement scaling and path geometry. The primary diagnostics are MSD and sinuosity; UD is included to show that spatial compatibility alone does not imply dynamic adequacy.

Scenario S2: incorrect number of states. The data-generating process will contain three states: an encamped state, an intermediate movement state, and a travelling state. Candidate models with one, two, three, and four states will be fitted. The purpose is to distinguish underfitting from overfitting. Models with too few states should show state-conditioned movement discrepancies,

distorted occupancy, and possible MSD or sinuosity failures. Models with too many states may generate redundant or poorly occupied regimes, which should appear in occupancy, transition, and state-conditioned diagnostics.

Scenario S3: missing time-varying transition structure. The data-generating process will use a two-state HMM-SSF in which the transition matrix depends on a cyclic temporal covariate, such as time of day. The fitted comparison will include a fixed-TPM model and the correct time-varying-TPM model. The goal is to test whether the diagnostics detect temporal organization in the state process that is not visible from the state-specific movement kernels alone. The primary diagnostics are occupancy through time, switching rate, residence time, transition counts, and MSD.

Scenario S4: misspecified movement kernel. The data-generating process will keep the correct state-transition and selection structure, but one state will have a movement distribution that is not available to the fitted simplified model. For example, the travelling state may have heavier-tailed step lengths or stronger directional persistence than the fitted kernel allows. The fitted comparison will include the simplified movement kernel and, if available, the correctly specified movement kernel. This scenario targets state-specific step-length discrepancies, turning-angle concentration, within-state MSD, and global MSD.

Scenario S5: misspecified habitat selection. The data-generating process will include state-specific habitat selection. For example, the encamped state may strongly select high values of the habitat covariate, while the travelling state may respond weakly or in the opposite direction. Three fitted models will be compared: a model with no habitat effect, a model with a shared habitat coefficient across states, and the correct state-specific model. This scenario tests whether the UD diagnostic and state-conditioned summaries can detect selection misspecification even when movement kernels and transition probabilities are otherwise adequate.

Scenario S6: misspecified barrier response. The data-generating process will include a linear barrier that affects movement differently by state. For example, the encamped state may rarely cross the barrier, while the travelling state may cross it with lower but nonzero resistance. Candidate fitted models will omit the barrier, include a common barrier effect, or include the correct state-specific barrier response. The primary diagnostic is the barrier-crossing statistic, interpreted jointly with state occupancy near the barrier and state-conditioned crossing opportunities.

Scenario S7: residence-time memory. The data-generating process will use an HSMM-like state sequence with non-geometric residence times, while keeping HMM-like state-specific movement and selection kernels. The fitted model will be an HMM-SSF, so the fitted state process will impose geometric residence times. This scenario is designed to answer whether the residence-time diagnostics can detect memory in the latent process. The expected signature is that residence-time and geometric-dwell diagnostics reject, while occupancy and average switching rate may remain compatible.

Scenario S8: state uncertainty and weak state separation. The data-generating process will be a correctly specified two-state HMM-SSF with weak separation between states. Two variants will be used: one where states differ mainly in movement and one where states differ mainly in habitat selection. The fitted model will be correct, but diagnostics will be computed using true simulated states, the Viterbi path, posterior state-path samples, and Viterbi-tube simulations. This scenario

tests whether state-process conclusions are robust to state uncertainty or depend on a single decoded path.

Reporting structure. The final simulation section should report calibration and power separately. Calibration will be assessed from S0 using the distribution of p-values across replicate data sets. Diagnostic sensitivity will be summarized from S1–S8 as rejection frequencies by scenario and diagnostic family. Each scenario should also include one representative diagnostic plot, chosen to show the mechanism of failure rather than only the numerical p-value. Before implementation, the remaining fixed choices are the number of replicate data sets, the trajectory lengths, the number of available steps, and the Monte Carlo simulation count used inside each diagnostic test.

7 Zebra model-comparison example

The empirical demonstration uses the zebra HMM-SSF example of [Klappstein et al. \(2023\)](#). Four candidate models are compared using Markov generation for all fitted models:

Table 5: Candidate models in the zebra comparison.

Label	States	Transition model	Interpretation
M1	1	Degenerate	Single-state SSF baseline
M2	2	Constant TPM	Two states, no time-of-day effect
M3	2	Time-varying TPM	Two states, time-of-day effect
M4	3	Constant TPM	Three-state fixed-transition alternative

The SSF formula in the zebra application is

$$\sim \text{step} + \log(\text{step}) + \cos(\text{angle}) + \text{veg},$$

which includes movement terms and a vegetation covariate. The time-varying transition model uses cyclic time-of-day terms,

$$\sim \cos(2\pi \cdot \text{tod}/24) + \sin(2\pi \cdot \text{tod}/24).$$

The comparison uses 99 Markov simulations per candidate model, giving a Monte Carlo p-value resolution of 0.01.

Table 6: Trajectory-level zebra diagnostics from the current empirical analysis. Values are observed discrepancies with Monte Carlo p-values in parentheses.

Model	UD Wasserstein	MSD	Sinuosity
M1 one-state SSF	19.04 (0.61)	1.25×10^6 (0.02)	0.175 (0.01)
M2 two-state fixed TPM	12.99 (0.30)	3.48×10^4 (0.01)	0.108 (0.05)
M3 two-state time TPM	12.81 (0.25)	3.66×10^4 (0.01)	0.107 (0.06)
M4 three-state fixed TPM	12.91 (0.24)	2.95×10^4 (0.01)	0.099 (0.13)

The UD discrepancy is not extreme for any candidate model in this run. MSD is extreme for every candidate model, including the multi-state alternatives. The one-state model fails strongly on

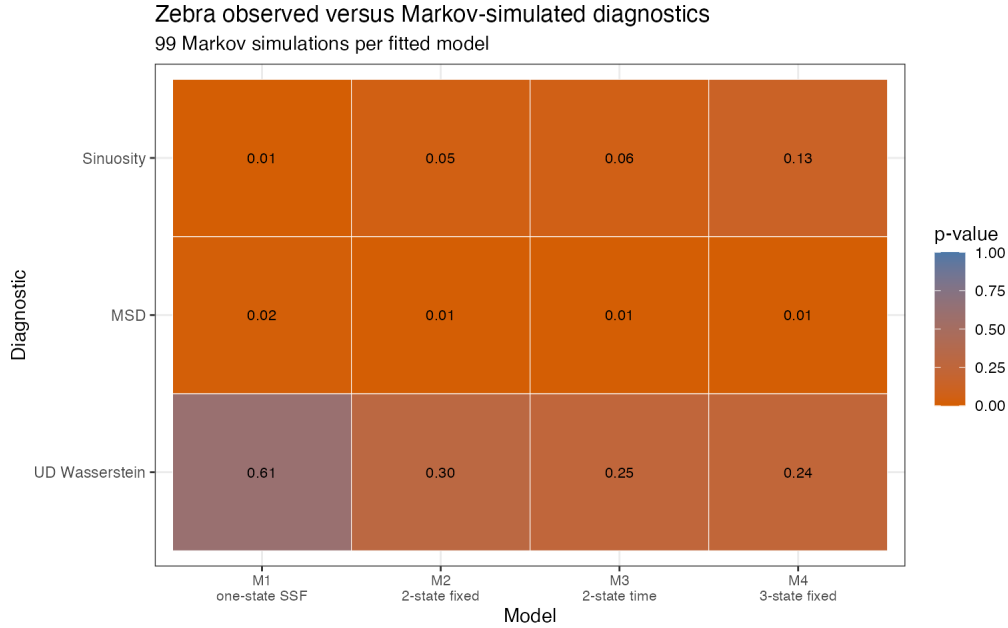


Figure 1: Monte Carlo p-values for the trajectory-level zebra diagnostics.

sinuosity, whereas the multi-state models are less discrepant, with the three-state fixed model giving the largest sinuosity p-value in this analysis.

State-level diagnostics are interpreted conditionally on the decoded observed state path. In two-state fits, per-state occupancy discrepancies can be identical because the occupancy proportions sum to one. The global occupancy discrepancy is therefore preferred for model-level reporting, with per-state rows used to identify which regime contributes to the mismatch.

References

- Avgar, T., Potts, J. R., Lewis, M. A., and Boyce, M. S. (2016). Integrated step selection analysis: bridging the gap between resource selection and animal movement. *Methods in Ecology and Evolution*, 7(5):619–630.
- Bastille-Rousseau, G., Potts, J. R., Yackulic, C. B., Frair, J. L., Ellington, E. H., and Blake, S. (2016). Flexible characterization of animal movement pattern using net squared displacement and a latent state model. *Movement Ecology*, 4(1):15.
- Benhamou, S. (2004). How to reliably estimate the tortuosity of an animal’s path. *Journal of Theoretical Biology*, 229(2):209–220.
- Besag, J. and Clifford, P. (1989). Generalized Monte Carlo significance tests. *Biometrika*, 76(4):633–642.
- Forester, J. D., Im, H. K., and Rathouz, P. J. (2009). Accounting for animal movement in estimation of resource selection functions: sampling and data analysis. *Ecology*, 90(12):3554–3565.
- Gelman, A., Meng, X.-L., and Stern, H. (1996). Posterior predictive assessment of model fitness via realized discrepancies. *Statistica Sinica*, 6(4):733–760.

- Klappstein, N. J., Thomas, L., and Michelot, T. (2023). Flexible hidden Markov models for behaviour-dependent habitat selection. *Movement Ecology*, 11(1):30.
- Langrock, R., Hopcraft, J. G. C., Blackwell, P. G., Goodall, V., King, R., Niu, M., Patterson, T. A., Pedersen, M. W., Skarin, A., and Schick, R. S. (2012). Flexible and practical modeling of animal telemetry data: hidden Markov models and extensions. *Ecology*, 93(11):2336–2342.
- Nicosia, A. (2026). Beyond the next step: A multi-criteria generative validation framework for step selection functions. *Methods in Ecology and Evolution*.
- Nicosia, A., Duchesne, T., Rivest, L.-P., and Fortin, D. (2017). A multi-state conditional logistic regression model for the analysis of animal movement. *The Annals of Applied Statistics*, 11(3):1537–1560.
- Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286.
- Turchin, P. (1998). *Quantitative analysis of movement: measuring and modeling population redistribution in animals*. Sinauer Associates.
- Villani, C. (2009). *Optimal transport: old and new*, volume 338. Springer.